Na and O abundances in NGC 5053

Owen Boberg, Eileen Friel, Enrico Vesperini

Indiana University

Abstract

NGC 5053 provides a unique environment to test our understanding of the lives of GCs. Recent studies have found that this cluster, along with NGC 5024 (M53), have interesting morphological features, such as a ~6-degree tidal stream related to NGC 5053 (Lauchner et al. 2010), and a possible, but still debated, bridge-like structure between the two GCs (Chun et al. 2010). These features suggest that the evolution of these clusters has been greatly affected by their gravitational interaction with the Galaxy, and possibly, each other. Using the WIYN-Hydra multi-object spectrograph, we have collected high quality (S/N ~75-90), medium-resolution spectra for red giant branch (RGB) stars in both clusters. Using these spectra we will be able to measure the Na and O abundances in ~20 RGB stars in NGC 5053 and ~100 RGB stars in M53. Our goal is to determine if the dynamic history of these GCs has had an effect on how they have processed material into multiple generations of stars. Here, we will present the Na and O abundances in NGC 5053.

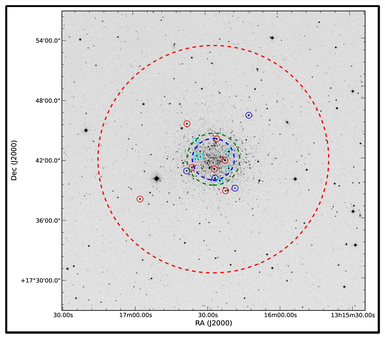

Figure 1: A DSS image of the cluster NGC 5053. The small circles mark the locations of the spectroscopic targets selected for this study. Their color scheme matches that in the legend of **Figure 2**. The blue dashed circle is the core radius, the green dashed circle is the half mass radius, and the red dashed circle is the King truncation radius.

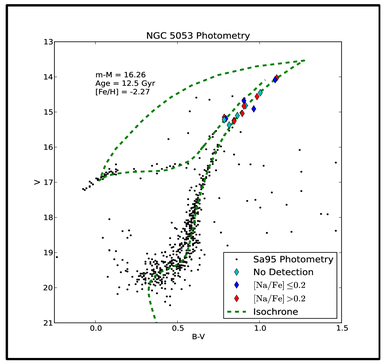

Figure 2: V vs. B-V color magnitude diagram for NGC 5053 taken from Sarajedini et al. 1995. The diamonds mark the stars observed in this study. They are colored according to the [Na/Fe] abundance we measured. No detection indicates that it was not possible to measure a Na abundance for that star. The green dashed line is a BaSTI isochrone with the parameters listed in the figure.

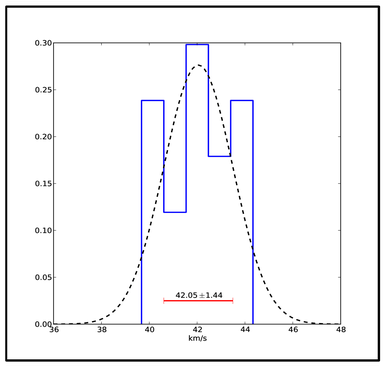

Figure 3: A histogram of the radial velocities of the stars observed in this study. The back dashed line is a normal fit to the radial velocity distribution. The red error bar is labeled with the mean and standard deviation of that normal fit.

Data

- Selection criteria:
  - V ≤ 15.5
  - Lie along the apparent RGB in V,B-V color magnitude diagram
  - Radial velocities consistent with the cluster
  - R_V this study: 42.05 +/- 1.44
  - R_V previous study: 44.0 +/- 4.9
- Medium resolution (R ~ 19,000) spectra were collected using WIYN Hydra
- Typical S/N ~ 75 – 90
- Exposure times of ~4.5 hours per Hydra configuration

Abstract Analysis

- T_{eff}, log(g), and V_t were determined using Sarajedini et al. 1995 BV photometry and 2MASS J,H,K
- Marcs models were generated based on the photometrically determined atmospheric parameters
- Fe and Ca abundances were determined using the *abfind* task in MOOG
- Na and O abundances were determined using the *synth* task in MOOG
- Fe and Ca abundances are consistent with other studies
- [Fe/H] = -2.29 +/- 0.08 (this study)
- [Fe/H] = -2.30 (Carretta et al. 2009a)
- [Ca/Fe] = 0.4 +/- 0.2

Table 1
Atmospheric Parameters and Abundances

ID	V	M_v	$(B-V)_0$	T_{eff} (K)	log(g) (cm/s^2)	V_t (km/s)	[Na/Fe]	[O/Fe]
2	14.03	-2.23	1.09	4460	0.99	1.63	0.5 ± 0.2	0.2 ± 0.2
3	14.08	-2.18	1.08	4400	0.98	1.66	0.0*	0.3 ± 0.2
4	14.45	-1.81	0.99	4540	1.21	1.57	···	0.6 ± 0.2
6	**14.56**	**-1.70**	**0.97**	**4590**	**1.28**	**1.53**	**0.5 ± 0.1**	**-0.2 ± 0.2**
9	14.68	-1.58	0.89	4660	1.37	1.48	0.0*	0.4 ± 0.2
11	14.83	-1.43	0.89	4570	1.38	1.54	0.9 ± 0.2	···
12	**14.91**	**-1.35**	**0.95**	**4490**	**1.36**	**1.59**	**0.2 ± 0.2**	**0.7 ± 0.1**
14	15.04	-1.22	0.88	4630	1.50	1.50	0.8 ± 0.3	···
15	15.10	-1.16	0.85	4640	1.53	1.49	···	···
16	15.15	-1.11	0.77	4760	1.61	1.41	0.7 ± 0.2	0.2*
17	15.19	-1.07	0.78	4750	1.62	1.41	0.0 ± 0.2	0.3 ± 0.1
18	15.22	-1.04	0.77	4860	1.69	1.34	···	···
19	15.24	-1.02	0.83	4650	1.59	1.48	···	···
20	15.25	-1.01	0.83	4910	1.73	1.30	0.7 ± 0.2	···
21	15.26	-1.00	0.83	4680	1.62	1.46	0.8 ± 0.3	···
23	15.37	-0.89	0.80	4700	1.67	1.44	···	···

Note.—* on a measurement indicates an upper limit

Spectral synthesis examples

- The spectra of stars 6 and 12 (listed in bold in Table 1) are plotted below
- These stars have nearly identical atmospheric parameters, but very different Na and O abundances
- Star 6 is Na rich and O poor: [Na/Fe] = 0.5 +/- 0.1 , [O/Fe] = -0.2 +/- 0.2
- Star 12 is Na poor and O rich: [Na/Fe] = 0.2 +/- 0.2, [O/Fe] = 0.7 +/- 0.1
- A good indication of the Na-O anti-correlation in the cluster

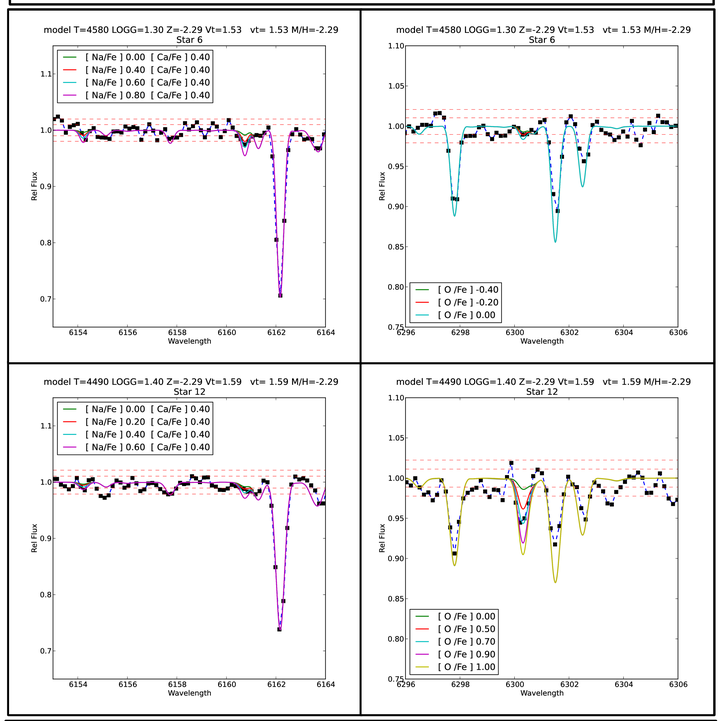

Figure 4: Two examples of the spectral synthesis performed to measure the Na and O abundances for this study. In each plot, the data are plotted as blue dashed lines and black squares. The red dashed lines in each plot represent the 1 and 2 sigma levels above and below the noise in each spectrum. The respective Na and O abundances used for each synthesis are labeled on the plots. For the Na synthesis, the Ca abundances was kept constant at 0.4. The spectral region with the 6300 [OI] line was corrected for telluric features and the corrected spectra are shown in this plot.

References and Acknowledgements

- Carreta et al., 2009a A&A,505,117
- Chun et al., 2010, AJ,139,606
- Gratton, Carretta,Bragaglia 2012, A&ARv, 20,50
- Lauchner et al., 2006, ApJL, 651,L33
- Smolinski et al., 2011, AJ, 141,89 (Radial velocity measurement)
- Sneden, 1973, PhD Thesis, Univ. Texas at Austin
- I am currently funded as an NSF Graduate Student Research Fellow

[Na/Fe] Distribution

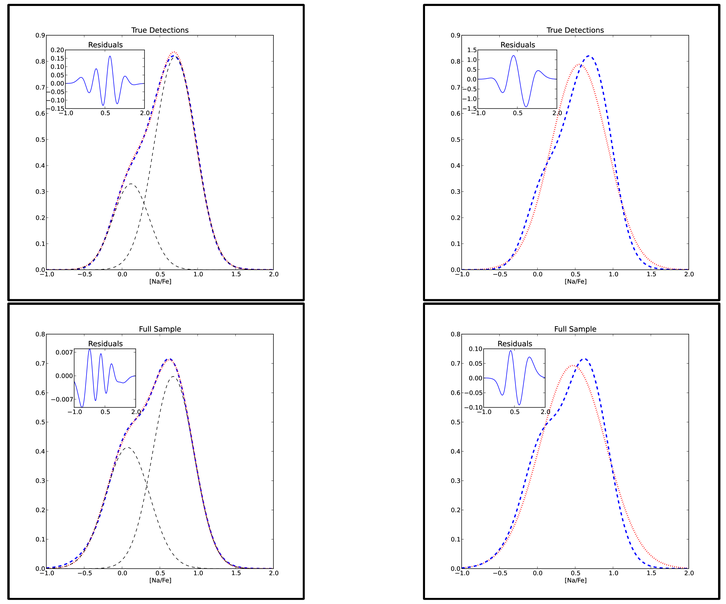

Figure 5: The [Na/Fe] distribution of the cluster members observed in this study. In each plot, the blue dashed line is a generalized histogram of the [Na/Fe] distribution. This was created by treating each measurement as a normal distribution with a standard deviation equal to the error on that measurement. The sum of these individual distributions produces the blue dashed line plotted here. The top row of the figure contains only those [Na/Fe] measurements that could be considered true detections (not upper limits). The bottom row contains the entire sample of [Na/Fe] measurements. For the upper limits, they were assumed to have an error equal to the standard deviation of the sample when constructing the generalized histogram. The red dashed line in each plot is a fit to the generalized histogram. For each case, true detections and the full sample, both multiple and single gaussian fits to the generalized histograms are presented.

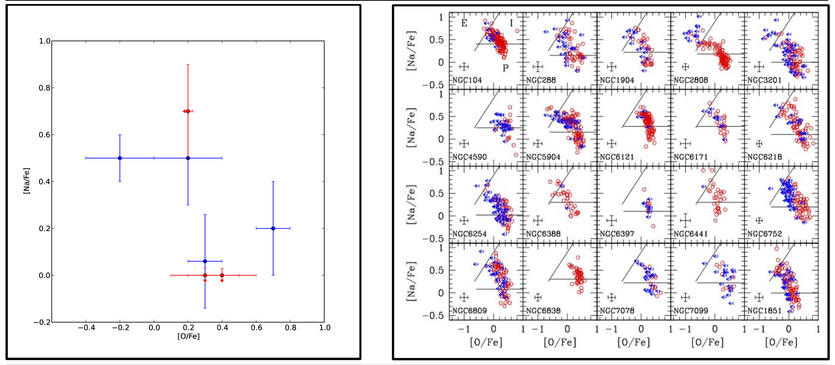

Figure 6: The left panel shows the plot of [Na/Fe] vs [O/Fe]. The points represented as arrows are upper limits. The right panel is taken from Gratton et al. 2012. This plot is provided to help put the results from this study into a broader context.

Conclusions

- The distribution of [Na/Fe] for our sample is not well represented by a single normal distribution.
- This [Na/Fe] distribution shows a wide range of abundances, spanning ~ 1.0 dex
- This range of abundances is larger than the typical error on the measurements
- Despite a limited sample, the Na-O anti-correlation for this cluster is consistent with that seen in other clusters.
- Provides evidence for multiple populations in NGC 5053.